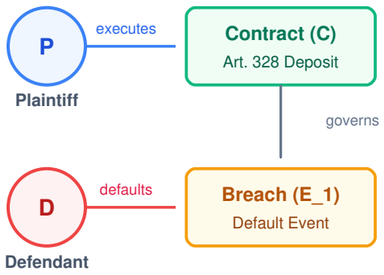
(a) Dynamic Case Graph

G(t) = (V, E, R, T, S)

Multi-Relational Topology

Bipartite Event Architecture



Node & Edge Embeddings

- Node features: $x_u \in \mathbb{R}^d$
- Edge relation: $e_{uv} \in \mathbb{R}^{d_r}$
- Timestamp: $t \in [0, T]$

Deontic Elapsed Time:

$$\Delta\tau = \tau(t_{filing}) - \tau(t_{breach})$$

Effective Procedural Duration

(b) Bochner Time Basis

Shift-Invariant Harmonic Space

Bochner Representation

Continuous Kernel Fourier Form:

$$\psi(\Delta\tau) = \int \exp(i\omega\Delta\tau)\, d\Lambda(\omega)$$

Theorem 4: Non-negative measure

$$\Phi_d(\Delta\tau) \in \mathbb{R}^{32}$$

16 Cosine + 16 Sine Frequencies

$\omega_k = 10^{-6(k-1)/15}$, $k=1..16$

Dual-Clock Continuum

Non-Autonomous Clock Integral:

$$\tau(t) = \int [1 - 1_{tolling}(\sigma)]\, d\sigma$$

Dynamic Horizon Integration

T_freeze (Art. 156 Tolling):

$d\tau/dt = 0$ during Force Majeure

R_restart (Art. 157 Reset):

$\tau(t^+) = 0$ upon Debt Acknowledgment

Statutory Limit (Art. 429):

Threshold: $\Delta\tau > 1{,}095$ days (3 yrs)

(c) Axiom-TGAT Attention

Continuous Relational Attention

Attention Mechanism

Temporal Kernel Projection:

$$q_u = W_Q\, h_u$$

$$k_{uv} = W_K\, [h_v \,\|\, \Phi_d(\Delta\tau) \,\|\, e_{uv}]$$

$$\alpha_{uv} = \text{Softmax}(q_u^T k_{uv} / \sqrt{d})$$

Multi-Head Stacks: H = 4 heads

$$m_u = \sum_{v} \alpha_{uv} (W_V h_v)$$

Aggregated Relational Context $\in \mathbb{R}^{128}$

Markovian State Gating

Contract Mutation Recurrence:

$$s_{k+1} = \sigma(W_s s_k + W_m m_u)$$

Automaton Latent State: $s \in \Delta^3$

Theorem 1: Banach Contraction

Lipshitz const: $\|W_s\|_2 < 1$

$$P(S_k) = \text{Softmax}(W_p s_k)$$

Predicts Valid / Voidable / Extinguished

(d) Normative DAG & TBox

Hierarchy Arbiter & Loss

Lex Specialis Arbiter (NH-DA)

Directed Normative Precedence:

Land Law 2024 (Art. 502) w=0.65

Civil Code 2015 (Art. 328) w=0.35

Lex Specialis derogat Generali

Specialized statute overrides general

Offline Satisfiability Matrix

HermiT SROIQ(D) Reasoner Mask:

$$M \in \{-\infty, 0\}^{4 \times 4}$$

Pre-computed Satisfiability Mask

Online Overhead: O(1) [16 ops, 8.15 ms]

$$L_{TBox} = \sum_{\phi \in T} \max(0, viol(\phi))$$

Subgradient Penalty: $\lambda_{onto} = 2.0$

Theorem 5: Subgradient Rate

Convergence: $O(1 / (\lambda \sqrt{T}))$

(e) Defeasible Adjudication

Dung Grounded Semantics

Dung Argument Graph AF

AF = ⟨Args, Defeat⟩

A_1: Claim
Art. 328 Forfeit

A_2: Defense
Verbal Waiver

UNDERCUT

A_3: Statute Bar (Art. 429)
$\Delta\tau > 1{,}095$ days $\Longrightarrow$ Procedural Bar

Theorem 2: Knaster-Tarski Fixed Point

$$lfp(\Gamma_{AF}) = \{ A_3 \}$$

Unique Minimal Grounded Extension

Adjudicated Decree

Masked Softmax Inference:

$$\hat{y} = \text{Softmax}(z + M_{k,:}) \in \Delta^3$$

$P(\text{Forfeit} \mid \Delta\tau > 1095) \equiv 0$

FINAL JUDICIAL DECREE

DISMISSED (TIME-BARRED)

100% Axiomatic Consistency Rate